Cardiovascular Disease

Prevention & Risk Assessment Guide

Major Risk Factors

Cardiovascular disease (CVD) — including heart attack, stroke, and other circulatory conditions — remains one of the leading causes of death worldwide. Many risk factors are modifiable, meaning lifestyle changes and medical management can meaningfully lower risk:

- **Hypertension** (high blood pressure) — forces the heart to work harder and damages artery walls over time
- **High cholesterol** — particularly elevated LDL and low HDL, contributes to plaque buildup in the arteries
- **Smoking and tobacco use** — damages blood vessels and significantly accelerates atherosclerosis
- **Diabetes** — high blood sugar damages blood vessels and raises cardiovascular risk independent of other factors
- **Obesity** — especially abdominal obesity, linked to higher blood pressure, cholesterol, and diabetes risk
- **Family history** — a close relative with early heart disease raises personal risk
- **Physical inactivity** and a diet high in saturated fat, sodium, and added sugar
- **Excessive alcohol use** and chronic stress
- **Age and sex** — risk rises with age; men are generally at higher risk earlier in life, while women's risk rises after menopause

Understanding Cholesterol Numbers

Cholesterol travels through the blood on different types of lipoproteins. Understanding the numbers on a lipid panel helps clarify cardiovascular risk:

- **Total Cholesterol** — desirable is under 200 mg/dL; 200–239 mg/dL is borderline high; 240 mg/dL and above is high
- **LDL ("bad" cholesterol)** — carries cholesterol into artery walls, where it can form plaque. Optimal is under 100 mg/dL; 100–129 near optimal; 130–159 borderline high; 160–189 high; 190+ very high
- **HDL ("good" cholesterol)** — helps remove excess cholesterol from the bloodstream. Below 40 mg/dL (men) or 50 mg/dL (women) is a risk factor; 60 mg/dL and above is considered protective
- **Triglycerides** — a type of fat in the blood. Normal is under 150 mg/dL; 150–199 borderline high; 200–499 high; 500+ very high, raising the risk of pancreatitis as well as heart disease

Blood Pressure Categories

Blood pressure is measured as systolic (pressure when the heart beats) over diastolic (pressure between beats). Current guidelines classify blood pressure as follows:

- **Normal:** below 120/80 mmHg
- **Elevated:** 120–129 systolic and below 80 diastolic
- **Hypertension Stage 1:** 130–139 systolic or 80–89 diastolic
- **Hypertension Stage 2:** 140+ systolic or 90+ diastolic
- **Hypertensive Crisis:** above 180 systolic and/or above 120 diastolic — requires immediate medical attention

Blood pressure should be checked at every healthcare visit, since hypertension often causes no symptoms and is only detected through measurement.

Lifestyle Modifications

The majority of cardiovascular risk can be addressed through sustained lifestyle change:

- **Diet** — the DASH diet (Dietary Approaches to Stop Hypertension), rich in fruits, vegetables, whole grains, and lean protein with reduced sodium, is well-studied for lowering blood pressure
- **Exercise** — at least 150 minutes per week of moderate aerobic activity (or 75 minutes vigorous), plus muscle-strengthening exercise at least twice a week
- **Weight management** — maintaining a healthy body weight and waist circumference reduces strain on the cardiovascular system
- **Limiting** sodium, saturated fat, trans fat, and added sugar intake
- **Limiting alcohol** and avoiding tobacco entirely — smoking cessation is one of the single most effective steps for reducing cardiovascular risk
- **Managing stress** and prioritizing adequate, consistent sleep

Warning Signs of Heart Attack and Stroke

Heart Attack

Common warning signs include chest pain, pressure, or tightness (which may radiate to the arm, jaw, neck, or back), shortness of breath, cold sweat, nausea, and lightheadedness. Women are more likely than men to experience atypical symptoms such as unusual fatigue, indigestion-like discomfort, or back pain, which can make heart attacks harder to recognize.

Stroke

The acronym FAST is a useful guide: Face drooping, Arm weakness, Speech difficulty, and Time to call emergency services. Other signs include sudden numbness (especially on one side of the body), confusion, trouble seeing, a severe headache with no known cause, or dizziness and loss of balance.

If any of these symptoms occur, call emergency medical services immediately rather than driving to the hospital — early treatment substantially improves outcomes for both heart attack and stroke.

Screening Recommendations by Age

- **Blood pressure:** at every healthcare visit, at minimum annually for all adults
- **Cholesterol:** baseline screening around age 20, repeated every 4–6 years for average-risk adults, more frequently with risk factors
- **Diabetes screening:** starting at age 35 for most adults (earlier with risk factors such as obesity or family history), repeated every 3 years if normal
- Those with a family history of early heart disease, existing risk factors, or a prior cardiac event typically need more frequent, individualized screening as determined by their physician

Medications

When lifestyle changes alone are not enough, several classes of medication are commonly used to manage cardiovascular risk:

- **Statins** — lower LDL cholesterol and reduce the risk of heart attack and stroke; among the most widely studied and prescribed cardiovascular medications

- **Antihypertensives** — classes such as ACE inhibitors, ARBs, beta-blockers, calcium channel blockers, and diuretics are used to manage blood pressure, often in combination
- **Antiplatelet therapy** — medications such as low-dose aspirin reduce the risk of blood clots in certain high-risk patients, though this must be guided by a physician given bleeding-risk considerations

Medication should always be prescribed and monitored by a qualified healthcare provider, based on individual risk factors and health history.